

Efficient Filtered Vector Search with Diverse Labels: A Trie-Free and Adaptive Routing Approach

I. XGBOOST ROUTER TRAINING

The main contents show that different LANNS queries favor different search paradigms. Based on this observation, ALPS uses an XGBoost-based router to select among three search paradigms: unified-graph-based search, proximity-graph-based search, and pre-filtering brute-force search. The router uses three query-level features, i.e., p_{pass} , $|F_{pass}|$, and $|L_q|$, which capture both vector selectivity and label-set structure. Moreover, Table I shows that all three features receive non-negligible importance scores on real datasets, indicating that each feature contributes to routing decisions. We implement the router with XGBoost because it can capture nonlinear feature interactions while maintaining low inference latency. In our implementation, the maximum tree depth is set to 6, the learning rate is set to 0.05, and the number of boosting rounds is set to 300.

We randomly generate 40,000 LANNS queries from all eight datasets, with 5,000 queries from each dataset. These queries exclude the test queries used in the main algorithmic evaluation. For each query, we evaluate the three candidate paradigms using the same implementations and parameter settings as in the main experiments. Specifically, the unified-graph-based paradigm uses TFNG with $\delta = 6$, the same construction parameter as UNG; the proximity-graph-based paradigm adopts the FAVOR-style search strategy used by ALPS; and the pre-filtering paradigm exhaustively computes distances over all qualified vectors, thus always achieving a recall of 1. For each paradigm, we follow the same search-queue-length tuning protocol as in the QPS-recall evaluation, varying the maximum search queue length from 10 to 70,000. We then assign the query to the paradigm with the highest QPS among those whose Recall@10 is at least 0.9. If multiple paradigms have the same QPS after measurement rounding, we break ties by first choosing the one with higher recall, and then by a fixed deterministic order: unified-graph-based search, proximity-graph-based search, and pre-filtering.

For router evaluation, we split 80% queries of each dataset for training and 20% for testing. The model is trained on the combined training queries from all eight datasets and evaluated separately on the test queries of each dataset. We also conduct a leave-one-dataset-out evaluation, where all queries from one dataset are removed from training and the trained model is tested on this held-out dataset. As shown in Tables 6 and 7 in the main contents, the routing accuracy remains above 90% under both settings, indicating that the learned routing rule generalizes across datasets. This accuracy is acceptable, given that the resulting ALPS outperforms state-of-the-art methods in main

TABLE I: Feature importance scores.

Data Name	Genome	Reviews	Amazon	VariousImg	Music	BookReviews	Tiktok	Laion
p_{pass}	0.9020	0.7522	0.7548	0.7635	0.9146	0.4902	0.6896	0.4038
$ F_{pass} $	0.0442	0.0985	0.1163	0.1963	0.0733	0.3567	0.2345	0.3912
$ L_q $	0.0538	0.1493	0.1289	0.0402	0.0121	0.1531	0.0759	0.2050

experiments.

As shown in Table 6, this training time cost is only a few seconds. In all algorithm-performance experiments, online latency is measured from receiving the query to returning top- K results, including feature extraction and model prediction. The feature-extraction and prediction overhead is only a few microseconds, showing that the router is suitable for real-time query processing.

II. CACHE HIT RATES COMPARISON

Frequent switching among search paradigms in ALPS may reduce cache locality and make repeated data accesses a major performance bottleneck. Therefore, ALPS+ groups queries with

TABLE II: Cache hit rates comparison (AP signifies ALPS).

Dataset	Alg.	L1 (%)	L2 (%)	L3 (%)	Dataset	Alg.	L1 (%)	L2 (%)	L3 (%)
Genome	AP	89.95	61.49	81.31	Music	AP	88.51	48.82	35.55
	AP+	89.99	83.80	91.66		AP+	90.53	58.61	73.98
Reviews	AP	84.16	44.68	39.38	BookReviews	AP	82.17	60.07	31.41
	AP+	87.76	54.35	77.44		AP+	88.67	60.71	74.63
Amazon	AP	88.79	56.14	27.64	Tiktok	AP	72.88	96.95	18.16
	AP+	89.84	61.57	64.11		AP+	73.44	98.04	57.04
VariousImg	AP	84.32	55.75	18.42	Laion	AP	80.88	59.43	49.95
	AP+	88.15	56.39	62.47		AP+	86.46	65.00	71.59

similar execution behavior to improve memory-access locality. Table II reports the cache hit rates of ALPS and ALPS+, where L1, L2, and L3 denote the three CPU cache levels. ALPS+ achieves higher hit rates at all three levels, indicating that query grouping effectively improves cache locality and contributes to its higher throughput.